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# Jeff Shen

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## Education

**H.B.Sc., University of Toronto** 2018-2022

Astronomy & Astrophysics, Statistics, Mathematics. Dean's List Scholar.

## Publications

3. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, J. Ted Mackereth, Yuan-Sen Ting, Jo Bovy. Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution, in prep.

2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. The Mass of the Milky Way from the H3 Survey, accepted to ApJ.

1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$. 2021, ApJ, 917, 79. 2105.11572.

## Research Experience

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|---|---|
| 2021 Sep - Present | **Narrowband upgrade for the Dragonfly Telephoto Array**<br>Supervisor: Prof. Roberto Abraham<br>Senior thesis. Containerized camera control software in C++ and Rust with real-time, multi-threaded data processing. |
| 2021 Jul - Present | **Constraining the strength of thermal tides with Stan**<br>Supervisor: Prof. Norman Murray<br>Independent research. Fitting dynamical models for the angular momentum of the Earth-Moon system to constrain the strength of thermal tides over geological time. |
| 2021 May - Present | **Predicting the ages of stars with machine learning**<br>Supervisor: Dr. Joshua Speagle<br>Independent research. Probabilistic predictions of stellar ages using normalizing flows. Quantifying the contribution of stellar/Galactic information to age estimates. |
| 2020 Aug - Present | **Estimating the mass of the Galaxy with Bayesian hierarchical models**<br>Supervisors: Prof. Gwendolyn Eadie, Prof. Norman Murray, Dr. Joshua Speagle<br>Undergraduate fellowship with CITA. Developed a hierarchical Bayesian model for modelling the mass of the Galaxy using a fast and scalable sampling algorithm. Collaborated with the H3 Survey team and presented results in ApJ. |
| 2020 May - 2021 May | **Molecular gas in high-redshift galaxies with ALMA**<br>Supervisor: Prof. Allison Man<br>Summer undergraduate research program with the Dunlap Institute. Analyzed ALMA data to infer molecular gas masses in several gravitationally lensed high-redshift galaxies. Presented results in a first-author publication in ApJ. |

## Honours and Awards

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|---|---|
| 2021 May - 2021 Aug | **Undergraduate Summer Research Fellowship, ($8,396 CAD)**<br>Canadian Institute for Theoretical Astrophysics, University of Toronto |
| 2020 May - 2020 Aug | **Summer Undergraduate Research Program Award ($9,500 CAD)**<br>Dunlap Institute for Astronomy and Astrophysics, University of Toronto |

## Presentations

2021 August **SDSS 2021 Collaboration Meeting, JHU/Online**
Lightning talk: "Predicting the ages of stars with machine learning."

2021 May **Stellar Stats Workshop, UofT/Online**
Lightning talk: "Estimating the mass distribution of the Milky Way with Bayesian multilevel models."

2021 May **Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online**
Poster presentation and lightning talk: "Molecular gas in a gravitationally lensed galaxy group at z = 2.9."

2021 Apr **Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online**
Talk: "Molecular gas in gravitationally lensed galaxies at z = 2.9."

2020 Oct **REQUIEM Galaxies Telecon, Online**
Talk: "Molecular gas in gravitationally lensed galaxies at z = 2.9."

2020 Aug **Summer Undergraduate Research Program Poster Session, UofT/Online**
Poster presentation: "Molecular gas in a gravitationally lensed galaxy group at z = 2.9." Awarded prize for one of top three posters.

## Independent Software Projects

4. Comprehensive scientific computing package for Rust: linear algebra, statistics, numerical optimization, linear models, and more. `github.com/al-jshen/compute`
3. Zero-dependency reverse-mode automatic differentiation package. `github.com/al-jshen/reverse`
2. Fast and easy random number generation in Rust with Wyrand. `github.com/al-jshen/alea`
1. Parallelized ray tracer for realistic scene rendering. `github.com/al-jshen/traycer`

## Media

2021 Jul **National Ratio Astronomy Organization (NRAO) eNews**
Digital newsletter feature: "Molecular gas in high redshift galaxies". https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas

2020 Jul **SURP Student of the Week Feature**
Interview: https://www.dunlap.utoronto.ca/surp-sow/sow12020/

## Outreach

2019 - 2021 **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - end of show.

## Volunteering

2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.

## Technical Skills

**Data**: Stan, Pyro, Pytorch, NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib, SQL, R
**Other**: C, C++, Rust, BLAS/LAPACK, Astropy, LaTeX, regex, bash, git, Docker, AWS, Javascript, React